

ActInf GuestStream 082.3 ~ "The Projective Wave Theory of Consciousness" Worden, Rudrauf, Williford

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hello and welcome everyone it's July 22nd 2024 this is active inference guest stream 82.3 the projective wave theory of Consciousness Robert Worden will give a presentation and we have several others to join and comment and give their insights so thank you everyone for joining and Robert please begin the presentation and also Robert close the YouTube on your side all good sorry I've it's coming through on my loud speaker which is a problem yeah it's all good like ready okay so this is a talk about the projective wave theory of Consciousness and there's already a paper about it on archive this month so you can look at the paper if you like but let me start by describing roughly the road map for the talk um the projective wave theory is very closely related to the projective conscious model PCM and we're fortunate to have three of the authors of that model on the call today but I'm going to give this talk in four different parts um the first part is talking about data and the idea is that any Theory of Consciousness has to fit the data has to fit some data in order for us to believe it and so the question is what kinds of data is there about Consciousness that the theory has to fit and I shall be arguing there the main kind of data we have is spatial Consciousness so that's part one of the Talk part two I shall talk about the projective Consciousness model PCM and to my mind the big virtue of the PCM is that it does predict account for just that the spatial nature of conscious and in that it's a great deal it does a great deal better than many neural theories of Consciousness and I shall probably pause at that point after Parts one and two and have discussion about the the first two points there but before that I think the PCM has great strengths but I think in common with all neural theories of Consciousness it has three problems and I shall talk about what those problems are but I shall emphasize that those problems are not specific to the PCM if you buy into the classical neurom model of the brain that neurons do everything then whatever your theory of Consciousness I think you're going to have these problems and so in order to get out of those problems we have to do something

that's not in the classical neural model that assumes that neurons in the brain do everything so we have to add something to the PCM and the something we have to add is there in three the assumption we're going to add to it is quite a big one it's an assumption that there is a wave in the brain which stores 3D spatial information in other words neurons don't do everything and then part four of the talk I shall bring those two and three together to a projective wave theory of Consciousness and talk a little about firstly how that solves the neural problems but secondly how it fits the data and some more General considerations about it so that is the root map for the talk so um we start off with what are the reasons to believe some Theory whether a theory of Consciousness or anything else why do we believe it basically we believe a theory because it fits data and it there's a complexity measure here that needs to fit more data which you can measure in bits if you like than the complexity of the assumptions you put into it if you just put in a huge number of assumptions and fit a small amount of data that's no use so there's a kind of basian balance between the assumptions you put into a theory and the data it fits and I've Illustrated the basian balance a pair of scales here and I've uh measured the the complexity of the theory and the amount of data it tests by a so-called zip test which simply says you describe these things in text files you zip them up and you see how big they are so established theories in science like the theories of physics or chemistry have a hugely positive base imbalance they have quite few assumptions and they account for huge amounts of data so how can we get to that point with the theory of Consciousness how can a theory of Consciousness produce a positive Bas imbalance between its assumptions and the data it accounts for so the question arises what is the data we have to test theories of Consciousness and there's a big problem which is well known that Consciousness is a private experience so there's not much public data about it but there are several kinds of data we can use such as data about levels of consciousness alertness and and being asleep and so on uh data about qual and now they are problematic because they're private phenomena but nevertheless you can two people can agree something about similarity scales of their qualia like different colors being similar or different and the form C is the shapes of things in spatial cons Consciousness and I shall argue that the main manifestation of Consciousness which we have what we're experiencing right now is a 3D spatial model of the world around us and that 3D model is very detailed and quite faithful to the real world around us and so that has a large information content um going down the list there's conscious imagination there is conscious of the self there are anomalies and illusions of Consciousness and there is the hard problem if you like the existence of Consciousness itself now I shall argue that point C the shape of spatial

Consciousness gives us much more data to test any Theory than any other type of data so what's the testable data in spatial Consciousness spatial Consciousness is like a 3D model of the real world around you that's what you're experiencing right now and it has a lot of the geometry of the real world and particularly straight lines in Consciousness correspond to straight lines in the real world and you can test this I won't go into the details of how so this agreement between our conscious experience which is quite information rich and the real world around us which is obviously information rich gives a lot of data to test any Theory Of Consciousness and I suggest roughly gives a thousands of bits even in a few seconds and I shall suggest that no other form of data about Consciousness comes anywhere close to this in terms of testable information content so if you really want to prove your theories right then it's got to account for the spatial form of Consciousness if you're going to believe it and down at the right here I put my contention that spatial Consciousness is if you like the element in the room Consciousness Theory so we have to consider how the brain does spatial cognition and all animal brains pretty much do that they have to understand 3D local space for a very fundamental reason as to control their every movement to move every movement of their limbs they have to understand 3D space around them and there's been huge selection pressure to do this well and we believe that animals do it very well even insects do it quite well and also it appears that most of the animal brain is devoted to that now if we assume a Bayesian hypothesis about cognition then the way animals do that is to build a maximum likelihood 3D model of local space in their brains from their sense data so it becomes fairly natural to expect that spatial Consciousness is going to be linked to this spatial model the question is is it a purely neural model as in classical neural models of the brain and if it is how does neuron firing represent 3D space and this is quite a difficult problem and really there are very few viable solutions to it but I shall switch now I shall put that problem on one side and switch to the projective Consciousness model PCM which has been developed by David and his collaborator since about 2017 and it's built on the free energy principle so it's based on a Bayesian model of local space in the brain and for neural for its neural process model it assumes the process model of the free energy principle and so the PCM assumes that the brain holds a 3D projective transform model of local visual space and the key thing is that PCM then predicts that spatial conscious Consciousness is like that model and this is a very good prediction it agrees very well the data for instance the trivial fact that straight lines in your Consciousness are straight lines in the real world so the PCM has a good Bas in Balance I should say this is very distinctive amongst many of Consciousness they very few of them

talk about the spatial nature of Consciousness and if they do they probably treat it a bit of an afterthought but the PCM puts it uh out in the front and I think that's a very positive feature of it and it has a good Basis in balance and it does make various other predictions which people have talked about in the papers makes predictions about illusions of Consciousness and makes predictions about predictive imagination and planning actions now I don't think those are quite as significant in the Basis in balance in the agreement with the data as the basic agreement with the form of Consciousness right now uh Consciousness from sense data but I shall say the PCM has three difficulties I shall emphasize they're not difficulties of the PCM alone I believe they are difficulties of any neural theory of of Consciousness and and I'll just list them briefly here because there going to be a slide describing them the first difficulty is neural selection how do we pick out just those neurons which cause Consciousness from all the others that don't and the second one is neural precision and speed we know that our conscious model of space is a very detailed and precise one can you do that with neuron firing I would suggest you can't and the third one is a complex decoding problem that uh anyway I'll come on to that so I'll describe these three in a minute but the emphasis is that they are not just problems of the PCM they are difficulties for any neural Theory Of Consciousness for instance IIT or Global neural neural workspace Theory and so on so what are the three difficulties the neural selection problem most neurons in the brain are not linked to Consciousness we know that from cortical lesions which don't interrupt Consciousness or the cerebellum or Beyond Mera is written quite extensively about this but neuron theories of Consciousness say that the activity of some neurons causes conscious ious and not others so the question is how are those special neurons selected and as far as I know there's no sound physical basis to pick out some neurons from all the others so you build in some selection assumptions into your hypothesis but those form a debit in your basian balance so they're a debit for any neural Theory Of Consciousness the second difficulty whoops uh all right second difficulty is how do neurons represent space now there are several possible ways of having neurons representing 3D space you can represent 2D space all right in neur can you return to sharing the the screen oh sorry I've lost sharing have I all good so um how do neurons represent 3D space there are several ways they might do it but they all have problems of precision and speed for instance there is a basin computational model of how animals build 3D internal model of space from their sense data and this is already on one of live the aii live streams at this YouTube address but that model shows that in order to build a 3D model of space even even for a small animal like a bee you need memory storage errors to be roughly the order 1% as small as that whereas if you model

neurons accurately properly they don't give you this Precision because in a neuron most theoretical models of neurons are ideal abstract neurons who out whose outputs are real variables of high precision and that's shown in the top of this picture here whereas actually neurons outputs are Spike trains which in the short time scale gives very limited Precision for instance in a fifth of the second possible time scale for Consciousness you may get about 10 pulses and that gives you a Precision of only 30% in that time scale so the Precision you get from neuro representations is very different from what you need to build a decent 3D model or what we experience in Consciousness and this I think hasn't been really spotted very much because most of the neural models that you see in the literature they don't model Spike trains it's enough to model networks of neurons and so they use abstract precise neurons so that's the second difficulty how can neurons store a precise and responsive fast model of 3D space like the one we experience finally uh what's happened here can you see that change third difficulty the third difficulty is what I call the hard triangle because conscious experience which you see in this picture here is a faithful hardly distorted model of the real world there's the real world A and there's Consciousness awareness C but what the models we have found in the brain are two-dimensional models and very distorted models and so they're many different models and they're highly complex encodings to get from A to B so if you want to get back from B to C you have to do very complex decoding to undo all the complex coding that most species brains do and so ABC is a hard triangle you have to have complex decoding assumptions in any neural Theory Of Consciousness and the worst thing of that maybe is that the decoding has to be species specific it can't be Universal decoding so those are the three difficulties for any neural Theory Of Consciousness including the PCM I believe so that's where we've got to I've covered topics one and two of this talk and now I think we'll break for any discussion or contributions thank you very much Robert for your presentation um up to now I can follow it very well I think it's uh you point out very important uh issues uh I should say and perhaps you know with scan that from the standpoint of the PCM even though indeed uh we have framed it in the context of active influence and more specifically the free energy principle um we do not necessarily uh um um until it does not necessarily until that we buy or we endorse or we believe in uh the current neural network models that are used in this context personally I've been a neuroscientist for many years done a lot of electrophysiology Legion work with human patient also I worked with rodent I've done a lot of functional Imaging and modeling and at some point in my career I thought that I did not understand the brain well enough and I stopped uh doing Neuroscience I stopped carrying quote unquote about the brain and uh for us

what is the most important in the PCM at this point is all you said is that we can capture some aspect of the phenomenology that seem essential etc etc so personally I have given up at this point uh trying to um devise or figure out a any you know realistic uh neural network models and when you take just one neuron as you said uh Robert it's a universe you just think about how receptors are up or down regulated through fast gene expression or dritic trees and their complexities and their geometry plays a role in computation I don't believe that the uh spiking uh of the of the neurons which we often call the output of the neuron are necessarily where we should be looking at uh and I don't know where we should be looking at I I committed a a a an odd papers a long time ago in 2015 uh in which I kind of said well I explain why I had given up hope at this point uh about brain science to account for that because I think we don't understand the brain I think there are principle we haven't figure out figured out yet by far and my um guess was that a bit like by analogy with computer science there are layers of abstraction mechanism of virtualization that hide the detail of the hardware uh which we don't understand at all uh and so so far it's a very open-ended thing so to me what matters the most about the PCM at this point is uh with respect to the brain it is neutral we don't know at this point I have absolutely no hypothesis except very gross anatomy hypothesis but or some you know possible details about you know but you write uh that's the hard problem and personally I have given up on trying to account for it thanks I found so much to agree with there and so yeah we we don't understand what's going on in the brain maybe neurons are doing something much more complicated than simple Spike trains but uh maybe there's also this wave any we'll talk about that Carl Ken do you have any comments at this stage uh Carl you want to go or I can if you don't mind I'll just follow up with I completely agree with what um deavid said um and I I will say my own sort of approach to the problem has always been sort of agnostic about the underlying realization so I've kind of started from all you know in my own thinking anyway always from the point of view of phenomenology with some kind of um you might say General acceptance of some sort of physicalism but agnostic about the details of that physicalism so agnos about whether it's um exactly what goes on in the brain or what level and so on um I I guess I've tended to even though I I agree you can't exclude it AR priori I've tended to kind of think well quantum mechanical theories are probably barking up the wrong tree but but that's just like my gut on that um and then personally sorry personally Ken at this point I'm so confused about the brain and about the physical nature of the universe that I'm even totally open that the whole thing is quantum I have no idea okay yeah as I said it's just sort of like my gut on that but not I can't really defend it in detail for a number of reasons um

owing to my own ignorance um but and I just add that uh and I I agree I think the way you presented the PCM generally was exactly right um you did mention that we do have something to say about imagination like action planning and about um anomalies and Illusions and and we you know we agree with you I think that the main set of data we aim to capture or the the spatial the spatial component of Consciousness then perspectival spatial component like you know the project what we use projective geometry to capture but then we do try to as you know try to derive certain other bits of parts of the relevant data from that basis so accounting for illusions um like the Moon Illusion and the Ames room illusion um and accounting for um imaginative action planning and also I should add some data about self-awareness so we have a paper on um self-awareness in this context as well um so yeah I think that's all I got so far but this is this has been very interesting so far right thanks again Carl on Carl gonna make any comments um no I'm just enjoying this as a spectator uh batting for both sides uh uh one comment it seems to me that we're focusing on the distinction between principles um about how the brain makes sense of its world and the process theories that attend those principles in terms of instantiation and there's a um an agnostic approach I think you can take in relation to process theories uh so I'm interested to see how much um Robert's Focus will be on the process Theory and the actual substrates which um you know I think is um interesting from the point of view of um actually simulating this kind of um inference um and also uh being able to reproduce it in silico in terms of you know create creating a a kind of artificial intelligence that has this this kind of competence and so uh there may be a more pragmatic Focus um I was intrigued by the previous slide um this this hard triangle um where you yeah just go to it there yeah uh this yeah it I I I didn't quite um resonate with this uh I think for a start the Notions of encoding and decoding for people like me would should be usefully unpacked um so everything in terms of belief updating is some kind of mapping or morphism or relationship um so I'm assuming mapping is a good word for it yeah okay so the world generates sensory inputs um and then the brain um represents those inputs I mean for B you just imagine you know the visual cortex with a highly distorted version of space in it all around the fvia and so on and all the other cortical maps are highly distorted representation of space and somehow you have to trammel those back into an undistorted representation so you've got to decode the encoding that's gone on I see so right so the conscious awareness is still part of the brain somewhere it must come from the brain somehow I think yes that we all I see so you're using encoding and decoding and a sort of uh distorting undistorting it's like a distorting mirror you got undistorted but I guess it relates to car distinction between neural

neural substrate and process Theory because indeed if you look at the classical sort of quote unquote 2D maps in the cerebral cortex Vis cortex with your retinotopy where you see a lot of deformation but also somato sensory ctis you think about the classical you know map of the body uh uh and you are ask saying well if that's the way it is kind of represented quote unquote at the level of that substrate then since it is deform and we don't experience it that way we should somehow have a process on top of that that use some morphism to transform this deformation I'm saying our Theory Of Consciousness has to tell us how that how that unscrambling Distortion happens yeah but I believe that the level of observation that we are considering here for instance those 2D quote unquote 2D because they are not really 2D but uh cortical Maps uh retinotopic or somatotopic or you know topic maps are a very quote unquote low level of processing uh which might be very far away from yeah what is going there might be some other super representation somewhere but we not found it yet yeah and it might not be visible at the level of the anatomy well yeah those are alternatives I agree okay so sorry Carl I interrupted but I wanted to make this precision too yeah no that clarifies it thank you but Carl I mean I think we can all be agnostic about certain things in the process model I think you and I at least are not agnostic about basm yeah I I would put sort of you know the notion of um sentience of the kind we're talking about as a process of basing belief updating I I put that at the sort of principle level sort of at the mechanics that that that doesn't commit to uh an infrastructure or a realization or a process that that realizes that uh um you know that kind of principle or that kind of uh mechanics um so I think the spatial wave theory is really a commitment to a process Theory um that could you know happily accommodate Basin belief updating and its particular application in the context of spatial perception and projective you know the projective Consciousness model you're muted um oh yes hear me good yes okay so moving on um so we've done one and two and what we're going to do in three is add something to the PCM and the thing going to add is not a little thing it's the Assumption the hypothesis that neurons don't do everything but there is a wave in the brain which stores 3D spatial information so that's what I'm going to talk about next so here's the hypothesis there is a wave in the brain that represents 3D space so it assumes there's some nearly spherical region of the brain with a diameter D and it holds waves which couples to neurons and and there are the sphere is Illustrated and a couple of waves wavelength Λ_2 that Green Wave and wavelength Λ_1 a shorter wavelength that red wave and those waves represent things in space and they represent the red thing R_1 and the green thing R_2 so there's a kind inverse relationship there the shorter the wavelength the longer the distance to the thing that's being

represented but obviously if it's a spherical region you can have waves in any direction so you can represent any direction and any distance up to a maximum which is defined by the minimum wavelength so that wave can represent lots of different things and the hypothesis is that that wave acts as a working memory for space it's really like a hologram now one issue with that is that if you want to represent objects at very large distances away then you get into wavelengths that's too short so that's why you may need a projective transform of space to pull in those long distances to be able to represent them on the brain on the wave by the way the evidence for this wave is going to be described in another uh YouTube live stream which we're making next week is going to be at that address so how well does a wave based spatial memory form it can represent approximately D of a λ cubed independent positions which is quite a lot and it can give you spatial Precision the order of λ over D and the point is that this can easily be 1% or better which is the Precision you need so there are significant benefits to wave storage it has high capacity you can put many objects in it it has high Precision it can be fast and it has low spatial Distortion it is just like a hologram and it's a very simple direct representation of space so a spoiler looking ahead to part four of the talk is that this is very much like spatial Consciousness question is there any evidence for such a wave I think there is evidence and there's three kinds of evidence first two comes from brain anatomy because there are spherical regions in the brain which are very well suited to hold this W in the mammals there is the thalamus and that's shown in green on the left of the picture and you can see it's approximately spheric and that is very suitably connected for holding spatial information it's connected to all the sense data and it's connected to all the bits of Cortex that work out what to do with the sense data and similarly in the insect brain there is a central body in the very center of it that is approximately spherical and it's very well preserved shape in all insect species so those are both pieces of evidence for a wave in the brain there is a the place for it uh so I believe there are three lines of evidence for a wave one is the computational Neuroscience which I've mentioned before that spatial cognition is a very important thing but neural memory for it is too imprecise and too slow whereas a wave storage model can possibly overcome these problems so that's the yellow evidence there evidence from the insect central body that is nearly round very Central very well connected and there's evidence from the Mammal Thalamus it's also round and well connected but there's also what I call a Smoking Gun for a wave in the thalamus that if you don't assume a wave the anatomy of the thalamus doesn't make sense but I won't go into that in this talk so together I believe these give quite strong support for the wave hypothesis in fact they give it a probability of 0.5 approximately

even if you try and wait the odds against it as I say you can see the discussion of this in another YouTube live stream we're about to make so then that's on from part three to part four of talk assume there is a wave in the brain which holds the near projective transform of 3D space and this wave is serving a very essential role to understand space to control all movements for any animal and the hypothesis simply is the wave is the origin of Consciousness that neurons have nothing to do with Consciousness at all it comes entirely from the wave this solves three problems the three problems that I outlined before firstly the neural selection problem you've no selection problem because neurons have to do with Consciousness the Precision and speed problem the wave has sufficient precision and speed to do it and the decoding problem the coding in the wave is just a Fourier transform so it's a very simple coding encoding and it's very simple decoding to give you back conscious experience so it has solved those three problems of neural theories but the main thing is that spatial for like the PCM preserving the Merit of the PCM the spatial form of Consciousness agrees very well with our conscious experience it has to have has to be projected because we need the project transform to pull in objects from very large distances to represent them by a minimum wave length so in summary uh the projective wave theory has a positive Bias imbalance doesn't have any Nares about it and it agrees very well with the main source of data about Consciousness so just some aspects of this Theory what is the physical nature of the wave a big objection might be we've never seen in wave because I believe uh it's probably not an electromagnetic wave or electromagnetic field and the main reason I believe that is that electromagnetic fields on their own don't store information for the time periods we need to store information and now electromagnetic fields might be part of the answer but I think they're not the whole answer and the big question then is why have we not detected this wave in brain if it's been there all these years uh I think the reason for that is we expect if there is a wave representing space we would expect it to evolve to have very low intensity so its energy assumption is very small and having that low intensity we simply haven't known how to look for it now coming to Quantum possibilities the wave might be a coherent Quantum excitation and there's increasing evidence that Quantum effects do have play a role in biology and it might even be a Bose condensate a BEC now those examples of those are superconductors and super fluids they're very rare state in nature but the key thing about a BEC is a superconductor can hold information for unlimited time so it would be good for memory and they're a very rare and distinctive state of matter and so if you assume that Consciousness comes from them you don't get to a trap of pan psychism or it might not be a Quantum effect because I think that this question is entirely so we'll set

that on one side and that's future exploration there's an issue about the evolution of Consciousness uh we know that Consciousness is our Consciousness is like the real world it's very like the real world and the question is why is this and you think well it must have evolved that way because there some need for it to be like the real real world but in neural theories there's a problem because neural computation in the brain the classical neural model is classical Dynamics not quantum physics and it's local and so concept is a non-physical non-local effect and they don't have any effect on neurons firing so the result is that Consciousness in a neural theory has no causal effect on Fitness that means there's no selection pressure on it to make it the way it is so in a neural Theory Of Consciousness we're driven to this slightly awkward conclusion that the special form of Consciousness that we have is just a bit of an accident that seems very unlikely whereas in the objective wave theory there is a good evolutionary reason Consciousness being like it is that is that 3D spatial cognition is very important wave is an essential part of it so the wave has been shaped by Evolution and it's been shaped to be just like the real world because that's the best way of controlling mve so Consciousness then is determined by the wave and that means Consciousness has evolved to be like so the wave theory solves another problem there finally the unity of Consciousness uh you may ask this question occasionally why do I only have one Consciousness and not several parallel consciousnesses and this also is something of a difficulty of neural theories because we have plenty of neurons in the brains have many parallel consciousnesses and brains can do some things in massive parallelism and if we had multiple parallel Consciousness it might be useful for instance of parallel planning of actions to see which one worked that's a difficulty for neural theories the wave theory doesn't have this difficulty because it accounts for the unity of Consciousness there's only one wave in the brain that is the source of Consciousness so there can only be one Consciousness I think I'm on to the last slide now pretty much the last slide if this theory is taken serious it makes Consciousness into a scientific problem rather than a philosophical problem because wave in the brain If a wave in the brain is the key to understanding Consciousness well if there's no wave then this theory is wrong get about it but there is a wave I think there are very strong reasons to link it to Consciousness because it has such a good fit to Consciousness so the exist whether or not the wave exists that's a scientific question it's not a question for philosophy and I think there's a good Prospect of answering it in the next 10 years so in 10 years time either this Theory will be the only theory of ousness or it'll be wrong and forgotten I think that's a virtue in a theory so um finally last slide on the hard problem David charma's hard problem the projected wave theory doesn't solve that problem

what I would say about that is all science has hard problems there is the hard problem of Consciousness why is conscious un link to matter there are hard problems in physics like what is space time or what is the most fundamental particle or the hard problem of cosmology why is there a universe so hard problems just don't have an answer scientists were like children keeping asking why say why this why that and we can go on asking why for as long as we like so for instance for fundamental particles we thought there was matter we thought there were atoms we got to protons we got to quarks and we keep going on and we'll never get the final answer but we still do science so I'm not too worried about the hard problem Consciousness uh so that is all and uh I've just laid out a few questions one might want to discuss in the remaining time maybe you could share your slide again so we can see the question if you have them on the slide uh sorry is it not sharing now I see the slide still um yeah so you can leave it maybe check your view David and then you can leave that up and we can have a panel discussion okay well thanks a lot um those who are here Ken and Carl and deid please feel free to uh comment and also people will write some questions in the chat and and I'll read one in if if you pause but anyone here who'd like to give a reflection or or any comments that would be awesome could I go ahead can you hear me yeah so uh if if I may I I I basically agree with what you say about the hard Pro just to take the very last thing that you mentioned about the hard problem and this is why earlier on when you said that um you had kind of a list of sort of um you know in your the idea of the basian balance for theories of Consciousness and you mentioned the existence of Consciousness as itself something that might be a problem for all neuronal theories of Consciousness I think you did that in on a slide two any Theory yeah yeah so I I agree I think it's a problem for any Theory Of Consciousness even the kind of um more metaphysical theories that people who reflect like Chalmers and others pansus who reflect on the hard problem a certain way end up with you can actually raise the same set of questions for them so there's really no progress there in my opinion so in other words if if you take the hard problem as a kind of indication that um well Consciousness couldn't be physical or or either that or the physical world itself must just be Consciousness which is kind of the two directions they they go there I think that's kind of misguided I mean we we have to sort of um accept that our ability to penetrate you know cognitively AR priori into nature of the cosmos is is limited so our the best we can do is make models and you know based on the phenomenology and try to arrive at a model that would give us some kind of neural uh or physical to speak more broadly uh physical um testable you know hypoth hypotheses we can test about the physical world what we think realizes Consciousness and it seems to me like

that's exactly what you're doing and that's what I would think is right too but I I did want to ask you um when you say that um no neurons are linked to Consciousness in your own model um do you mean maybe I just don't understand exactly but then they're not linked directly to the Consciousness NE but they're involved in the realization of of the wave is that oh they are abely abely they're controlling the wave they're setting up the wave they're all around the cortex they're using sense data to to fix the wave and it's the wave that makes Consciousness okay and on the issue of unity well so suppose we identify this type of wave um I mean it's at least you know conceivable that we would find multiple such waves would you concl conceivable I think the evidence in the thalamus and the central body is one wave okay and I think one wave is what you need to understand one space around you okay so but if you found multiple such waves yeah absolutely that would be an issue yeah okay I mean would you conclude that there must be multiple consciousnesses then in a single brain or would you would you think that the theory is wrong no I would have a selection problem just like neuron theories a selection problem selecting neurons I'd have a problem selecting which wave was conscious I see H rather than conclude that there are multiple you want to avoid that and my experience is I don't have multiple conscience yes my right but that might just be because only one of them gets to report yeah absolutely you know like like the way they you know I know it's debatable but the way they um you know often interpreted the the split brain uh cases coming out of yeah good question good question I mean I say one wave but we have two thalami so the Assumption there is that those waves are linked and perhaps in split brain they don't get linked H okay I think two thalami have one wave in them but that's a deto I see and then when it comes to the kind of wave like so you you don't think it's an electromagnetic wave uh you you you think it might be a Bose Einstein condens it that's that's the hypothesis I think that Karen Smith once uh Flo it's been around for some years yeah Alexander Karen Smith in a and the book uh that's the yeah not not the book on the origin the famous book on the origin of life but he wrote a book on Consciousness back in the 9s I think um yeah there was Michael Lockwood as well in the 80s I think it's a long history okay so H how um when you say that you think we will'll be able to probe this within the next 10 years I mean could you sort of sketch how that might go how we might be able yeah I mean just this is not this is probing the idea of the wave right as I say I think there are those three pieces evidence there's neural models that you can't build decent neural models there is the thalamus and there is the insect central body I think we can go further on each one of those we can test them we can test the thalamus ideas much better using connectome data we can test the central body using

connectome data and we can test the computational models by just trying to build them and see if we get in so there is science to be done to falsify this hypothesis and it may succeed in falsifying it okay um but how would you measure uh such condensate I mean if you want to be able to demonstrate the existence prop ideally you'd like to actually directly detect it I I think it's a bit like when poy in the 1930s predicted the neutrino he thought it absolutely impossible to detect neutrinos and it took 25 years to find it and I think it'll take a long time to really find this wave in the brain because it's very low intensity probably and because we just don't know how to look for it and I think we have to do the biophysics until we had a convincing case for I this kind of wave then we can build a detector but until then I think we won't detect it like neutrino or like dark matter similar thing predicted theoretically years ago but still not detected I'll read a question from the live chat okay Andrea Val wrote thanks for sharing ing such interesting ideas I wonder how the PCM or 3D model can explain or deal with the un/consciousness of emotions so where is affect and such in this kind of spatialized model yeah so you want to do in the PM first David uh if it's okay Robert maybe you have read some of recent work on that but we we we we I have a specific answer to that question if you yeah you go first yeah sorry about that uh well one things that uh we realized working with that particular geometry is it transform information in the sense of information in space you can think about distribution in space for instance that are going to be changed by projective transformation uh in a manner that affect directly entropy so if something is very far you know in you have some uncertainty about the localization of these things so there is a probability function that represent that uncertainty about localization of that thing uh uh and there is only one thing in your field of Consciousness to simplify the problem uh then the entropy of the whole Space if you if you consider the whole 3D space uh and some scalar on it that would represent probabilities uh is uh uh um is very small whereas uh when you get closer for instance by imagination through projective transformation that anticipate action you get the object closer and you transform also the information the the variance if you wish of your distribution so it become broader so the entropy increase actually as an effect of projective transformation of the on the distribution of information so how does that relate to emotion what we uh realized is that uh um it's can account for some drives that are related to exploration or exploratory Behavior or search Behavior to reduce reduce uncertainty or maximize information uh and that just does that so basically the projective transformation uh uh by in virtue of making probabilities function in space uh change their shape as a function of perspective taking if you wish until or induce a drive to approach or avoid things or in in in

this context to approach uh to have be approach Behavior either through imagination or through actual Behavior because that will maximize information by increasing entropy of the object and you have more details so that's something that we wrote a paper that is currently under review in biological cybernetics about that with car with sorry gregar Ser Pera first auor with a bright mathematician and we can do a demonstration and actually derived theorems about that and also simulations that show interesting behaviors that depend on projective parameters that will show that the behavior of the agent uh depending on the parameter will be more focused so they will tend to always approach one object because they're driven by this anticipation of increasing information through projective transformation uh but uh uh uh with other parameters or values of the parameters on the contrary they start now oscillating between objects they start losing focus as if they were kind of ADHD uh so there is a deep link between let's say um the evaluation of space the appraisal of space uh um the drives that are uh uh that unfold from this information and the behaviors that uh you have for instance approaching uh uh uh and that geometry that very geometry we are talking about yield that type of thing so here we are talking about what is typically described as epistemic emotion basically curiosity so we have Natural Curiosity and part of it that drives behaviors is a result of the very geometry that represent information and appra it according to that theory we I cannot enter into the detail on that but you can also look on papers that we wrote In what was that 2022 in uh uh the Journal of theoretical biology in which we use little robots uh to demonstrate adaptive and maladaptive behaviors uh with principles also derived from the geometry and the free energy principle or something equivalent uh uh to get a wealth of affective behavior of approach avoidance but also uh uh social cognition you know you kind of we can kind of simulate social anxiety um we can we obtain joint attention Behavior as a result of mutual perspective taking uh and there are lot of things we could account uh that we can use in this model in in this framework to account for emotion there is a lot of work ahead but we believe that there is and we have good reason to believe there is a deep link between space and emotion that's actually called Aesthetics that Consciousness is a form of Aesthetics where space and emotion are uh untangled and that we can fairly well uh account for many phenomena that relate to emotion and behavior through this model but that's sorry for the long talk that's great should I give my my so answer is very much along the same lines that emotion and space are deeply linked um in the wave model I mean the project the the model of space contains all of space the external space and the internal space of the body and an important part of emotions is feelings in the body and they are part of your conscious awareness part of

the uh the wave model objective model just like the others and so basically this the PW Theory and I think the PCM says all Consciousness is spatial we have qualia but they exist in space you hear you hear words you experience colors you experience all of those and you experience feelings all of them in space so if you look at all the components of emotion human emotion for instance the words we say to ourselves and the feelings we have in our bodies they are all represented in that comp that conscious space so human emotion is a complex story which perhaps we don't want to go into but basically it comes down to the way we experience it is in that model of space and if I may I would have that if you take the standpoint of pure phenomenology and you ask yourself when I experience an emotion for instance anxiety or fear something very intense that is very sent uh you can ask yourself how much of that phenomenology of that experience is actually quote unquote reducible to some spatial experience things for instance you know the experience of the body with the heart that starts P pounding in your chest and you feel it stronger your lungs that change there there you know you breathe faster for instance and you can feel that movement in your body and all of that is special what you feel in your your gut or your intestine get kind of stretch or whatever you we have metaphors but actually a lot of the details of the phenomenology of emotion of feelings let's say uh I believe is actually spatial but it's overlooked people don't think about it but in particular think about bodily awareness there is a lot of special stuff going on here yeah I agree um and um mindful meditation talk a lot about space and talk a lot about the body I think they put emotion in the right place Carl or Ken want to add anything or I can look for another question in the live [Music] chat well I I have uh one question um so I mean your current hypothesis is that this wave is realized in the thalamus is that is that right that's right yeah um okay so that sort of um would you know there's this kind of debate about whether or not Consciousness is cortical or or subcortical so this would definitely be on the the sub I mean I think there's a lot of sense in the global workspace Theory and they were talking originally about a workspace involving the thalamus and the cortex mhm I think it's it the Consciousness really comes from the thalamus integrating all that spatial information yeah so it is down so what the the follow-up uh but yeah that's fine with me uh frankly but the follow-up question is um well it's not exactly a followup to that it's more a general question about your way of thinking about it how I mean I know maybe you can't really answer this until we have a more specific hypothesis about exactly what kind of wave it is but could you say something generally about how the wave um integrates with or or what's what's the right way to put it use recruits and you know the kind of computational apparatus of the brain so that you know you

end up you decide to uh oh I'm going to reach out you know yay far and grab this this bottle how does the um how does you know computationally speaking how does the wave integrate yeah I think that's a very good question and could have a very long answer but basically um The Wave integrates information from all the sense sta so from your hands from your ears from your eyes all that information goes through the cortex and cortex decides where it's coming from and broadcasts signals back to the thalamus and the thalamus wave integrates all those signals together produce the basian best model of what's happening at this moment based on touch Pro perception hearing Sound hearing vision and so on so the computation it's an integration of many sources of information uh possibly in a cycle between the thalamus and the cortex okay but if if you need to um you know for example program and action so you you that's kind of the you got your your sensory um the integration of all your sensory information now you need to program an action how do and maybe this is just my ignorance how how do waves how can waves be used in a way to well generate programing an action and then give it back to your your neurons I mean I've only got beginnings of answers to that but basically wave is very good for representing 3D geometry and for calculating displacements I know where my hand is I know my coffee cup is so the wave helps me calculate the displacement which way I have to move my hand to pick up the coffee cup so it's that kind of spatial calculation that the wave is always helping us do just to move our limbs um I have a followup question on that if I may can do you mind uh yeah no go ahead go ahead you invoked earlier F transform and inverse F transforms so at least 3D for transform and inverse for transform uh in order to do the encoding and decoding uh to you know go move from the wave to the neural substrates back and forth um of course as you know uh there is a limits in resolution uh when you do a forier transform uh you have the uncertainty principle uh that rules the absolutely faal resolution that you we can have of course it depends on the number of sample I mean there are a lot of arithmetics behind it you know you can be more more and more precise as you have more and more bits of uncoding if you wish um so uh you you say that you know neural classical neural theories lack build upon substrate that lack enough Precision uh in spatial resolution for instance to account for our experience would you think that in the context of a wave theory the type of uncertainty principle that are attached to it could cause a problem of resolution too or that it's the the granular there is a problem yeah there is a problem of resolution I alluded to it in the slides that is that if you got a minimum wavelength Λ and a diameter of the sphere D the maximum Precision is Λ over D and that can't be infinite Precision so that's the uncertainty principle but the idea of the FIA transform is not that

something the brain does a Foria transform it is the fundamental bridging law for Consciousness does the Foria transform it says that Consciousness is a Foria transform of the way full stop and for transform is there in fundamental laws of nature like like Quantum Mechanics for instance so it's saying this is a fundamental law of nature just like quantum mechanics says momentum is a for a transform position so the the bridging law for Consciousness says the form of Consciousness is a for transform of the way I'll make an a short comment or Carl do you want to after you okay yeah just if so thank you for the presentation and it's it's really interesting on on many levels first year you and the the paper goes into more detail on this you lay out what your priors are and kind of the dependency structure to actually reach at least an estimate of probabilities which is it's it's really interesting it gives a lot of opportunities for people to dig in and say yeah I think that's 10% likelier no it's a zero this is not possible and kind of close off a branch of inquiry um and also this discussion here about uh the algorithmic and the substrate realiz ation interaction with the substrate um and even this poly computation idea of of the observer's model of what's happening in a process and kind of like the dark matter there like The Observer might suggest that there's a cryptographic process though they couldn't decode it and then that would have an entropy distribution that that wouldn't be distinguishable in the absence of the decryption from a noise but it would be like still a hypothesis kept alive and you could say I think it's a 1% chance of this being the case um one one thought is how high Precision really is the spatial generative model in that there could be visual upscaling we know there's attention play to make the periphery look sharp and have color vision yeah and so it's possible that especially for like distant objects small objects that that they could be bundled and just seen as a assembly a larger object if they're touching like the car body and the wheel and then and then the same way that in the periphery color and visual attention seem to give this extreme like wow that's like exactly this angle if you were to go over there with a protractor but that could be as we're sort of seeing with diffusion and generative AI models that could be quite a low resolution structural inference followed by a just compatible visual realization that that ALS and then just last comment really interesting to explore how does evolution and intergenerational change shape the substrate that the sense data wash over so that non-stationary stable States whether they're rate encoding of neuron or G Electro and all these other even potentially more exotic options however those stabilized wow you you I mean th those are real questions to investigate particularly the Precision of our conscious experience and how much is quotes an illusion but I personally think that if you look at something which is a straight line you

experience it as a straight line with some kind of precision that's of the order of 1% a deviation of 1% from a straight line would show and so that's a lower bound on the Precision of your model but again there can be lots of discussion about this yeah interesting Carl any thoughts well just a few sort of um practical and specific questions um so the nature of this wave if I wanted to um model it I mean are we talking about a standing wave are we talking about a soliton are we talking about spherical harmonics on a on the surface of the um the thalamus are we talking about you what kind of basis set might I might might I use for for this kind of wave even before asking questions about what what the wave amplitude actually means what so what what's the functional mathematical form of this way a very good questions whether it's a standing wave or a moving wave I don't know the nice thing about a Bose Einstein condensate is that it acts like a longlived quantum mechanical wave function in other words it has phase which can be detected and it preserves phase for long times and the wave might have to preserve phase in order to represent positions if you have a standing wave like $\cos \Theta$ in some direction in this sphere then there's an ambiguity it represents plus Direction and the minus Direction and what happens to the minus Direction so I would rather have phase in the wave like a Bose Einstein condensate but they are very deep questions and I don't need the answers I mean we could start modeling and and see what happens just propose some answers uh basically you just need a spherical volume that holds a wave excitation with a certain maximum wave vector and we could try out standing waves moving waves and so on I me what would um differentiate um a condensate from a formulation in terms of a Time dependent scrolling a wave equation um well I think your time dependence Shing a w as as everybody knows the problem with Quantum in the brain is that decoherent smashes it in 10us 20 seconds but that doesn't happen to a bone condensate a bone condensate is a long lived Quantum State and so that's you know an ordinary Shing equation decoherence is going to smash you in no time at all unless you have some incredibly clever insulation like in a quantum computer right but I mean you know lifting from the microscopic to the macroscopic I mean that that would suggest that you you need very very low temperatures thermodynamically speaking I mean you nobody just observed a BC at room temperature yet but we've got close and the idea is that nature might be clever than we are one angle perhaps from the quantum Fe from from Carl and Chris fields and others is and so what I heard in the qu in the basis set question Carl was um this brain region that hosts the functionality or or a core aspect of the functionality like one option is it is like the the the wave is represented inside of that sphere like just a there's probably multiple way even that could happen but it's

like a little reconstruction wave but symbolized another option would be like a holograph the representation of space is encoded on the interface of that region with other regions and so that kind of surface is maintained and stabilized and has information flowing through it and then there wouldn't necessarily be some in the middle but it'd be more like a ripple on a Surface which was the the Bas ass set question Daniel that is very interesting because in the thalamus there is the thalamic reticular nucleus TRN which is a thin skin around the surface of thalam and you could just say that is holding the surface wave and the wave is projected inside from the surface it would be useful um if um to engage Chris Fields here there are two very distinct uses of the word holographic here you could be referring to the holographic principle yeah um whereby all the information on the bulk on the inside say of the thalamus um was um on the surface on the you know the reticular nucleus of the thalamus for example or just on the surface um you know in in one sense and um and that is what the rest of the brain sees and possibly Consciousness sees um as AO but that's a very different use of the the notion uh of um a holographic projection using um um constructive and D active interference to to encode things um so then you're absolutely right my my my question was really searching your um to distinguish are we talking about um a wave function um in the phala so are we talking about um application of the holographic principle to um the classical information on the surface of some bulk that was the thalamus and you could then argue if you if you wanted to that the bulk which can never be observed from the outside unless you are the phalus you never know what's happening on the inside um you could only observe what's what's on the surface um um you know that would be a different applica a different notion of of the of of a wave theory that would um certainly be consistent with Quantum information Theory and also um consistent with um um Chris Field's inner screen hypothesis that could be spatialized and endowed with with um a in virtue of the fact that the um on the holographic screen itself on the on the surface there can only be classical information and that information can now be um cast in terms of classical basing inference um but it also um relates to repeat to Chris Field's inner screen hypothesis um which is um you know if you like taking Quantum information theoretic principles uh and trying to apply them to uh to Consciousness that eludes the problem of the of applying quantum mechanics that is only relevant at the very small scale or or for very precise or low temperature systems uh you could if you want to apply that kind of maths to um macroscopic phenomena then you have to apply it in a scale invariant context and I think Quantum information Theory would C would certainly do that so D yes I was exactly sort of trying a probe are we is the is PW red um the the um Bose Einstein cond condensate

variant of or alternative version of uh the projective wave is that committed to Quantum Mechanics or is it really um an um an expression of um condensate like Dynamics in the context of quantum information Theory uh and if it is does it live on a holographic screen or does it live in the bulk and if it lives in the bulk what what are these wave functions over so these are the S practical questions I was I was trying to get very interesting questions and it's all to play for really I I have an open mind about them but um my feeling is that the becc is a way of getting past this decoherence argument from tmar and others that you know neurons have to be classical because of and the BC it it does work very well as a memory that's that's the Practical point of it and working memory I think is what you need for space awesome um perhaps each can give a last round or thought like where where do they feel this takes them or what are they looking forward to and uh Ken then David and then Carl and then Robert I yeah I don't have anything new to add just that I find this really interesting and I remember um reading uh Karen Smith's book kind of arguing that Bose Bose Einstein condensates might be might be um important for understanding Consciousness and thinking that among the quantum mechanical hypotheses I like that one the best so praise but well as I said earlier I just sort of have this kind of bias but I'm totally willing to be talked out of it um and uh yeah the the reason I mean I don't want to go into the details but it kind I think Karen Smith's concern was about the unity of Consciousness primarily and um also kind of made the case that yeah as you were suggesting that maybe um maybe as you put it nature is clever than we are and there's some way to realize that Bose Ein sign condens it in the brain without it you know without the decoherence uh issue but anyway uh I'm just to say this is all very um very fascinating to me and and I am am happy to continue the discussion next week right isn't it next week yeah thank you Ken and for my part yes also looking forward to continuations of the discussion you know my um my direction of questioning will be um something on the lines I think both kenth and David alluded to which is the um the link between spatial waves in the brain and the projective geometry that they speak to and the information geometry that underwrites um belief updating and the basing mechanics that that um inherit from from the free energy principle U be it classical mechanic classical information processing or um the you know looking at um using Quantum information Theory to you know to read um read belief updating I'm not sure I'm going to get any answers but that's what I'd be interested in hearing more about and the process series we haven't talked about Mortal computation yet so we need to talk about Mortal computation oh yes David any last thoughts well there sort of a of of a man who is 50 now and feels that the more he learns the less he knows so was predicted in all

the books I had read in my years uh and so I'm whole open to that complex Universe I certainly believe that nature is clever than I am for sure and I find that fascinating I I like the rational it's not like you know there is a lot of stuff about quantum mechanics and the brain and Consciousness that to me has always send Wishy wasy if not completely Waco uh uh but uh I came back a bit from that I had a lot of like can uh priors negative priors about it uh I remember once in Tucson Arizona at the famous conference toward about the science of Consciousness there were this bunch of guys who called themselves they were German the quantum Consciousness Academy and they were all dressed in black and they all you know headset and microphone like that and they were like speaking nonsense sense as like a cult but I found that very fascinated because I I I used to like very fascinating sorry I used to like this conference back in the day when it was very psychedelic uh but that was kind of you know and I come back to that because my ignorance I'm becoming aware more and more of my own ignorance uh I said well why not and definitely I think that maybe Carl can try to to to do that if we could get Chrisfield in the next call he would certainly be among us the most competent uh I tried to get Gregar Ser per uh to the call but he was not viable uh he's a very bright mathematician and he Master quantum mechanics like like if it was you know I don't know it's nothing for him or Daniel Bena would be the best also with Gregar but I'm afraid I won't be able to have them here but I think it would be very useful to have some people like that who definitely know more than me about those things and Chris might be interested to join I don't know car but looking forward to the the next uh uh to the next discussion one correction they were Swiss David they were they weren't David they weren't they weren't what they were Swiss they were Swiss Men In Black I think yes they kind of look like people from The Matrix like oh they were Swiss they were not German they were Swiss yeah and they had they had a leader they were always in contact with Super H wow okay word yeah please Robert and and and we'll contact Chris but go for the all your contribution that that's been really fascinating I think the virtue of this theory is that it is falsifiable in other words if there's no wave in the brain forget it this theory is wrong and that's exactly the question we're going to be looking at next week I look forward to having your contributions awesome thank you all see you at 16 UTC on July 29th okay thank you thank you bye for